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A TabPFN-Based Framework for Credit Risk Prediction in Automotive Green Supply Chain Finance

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Abstract

As the automotive industry undergoes a green transformation, digital upgrading, and increasingly intensive supply chain collaboration, the supply chain finance credit risks faced by small and medium-sized enterprises (SMEs) in the sector exhibit characteristics such as multi-source interaction, nonlinear transmission, and class imbalance. This study uses 210 SMEs in China's A-share automotive sector from 2020 to 2024 and constructs a credit risk evaluation system covering 56 indicators across the macro environment, financing enterprises, supply chain characteristics, and core enterprise credit support. Methodologically, DE-LightGBM is employed for feature selection to reduce redundancy and noise, while TabPFGen is introduced to generate synthetic risk-class samples. Business logic constraints and a Nearest Neighbor Distance Ratio filtering mechanism are further applied to improve the plausibility and fidelity of generated samples. Empirical results show that the TabPFN model achieves superior predictive performance after feature selection and data augmentation, and the Wilcoxon signed-rank test confirms the effectiveness and stability of sample augmentation. In addition, the ablation experiment demonstrates that green-related features provide significant incremental predictive value for supply chain finance credit risk identification. The proposed framework provides a useful reference for SME credit assessment, risk early warning, and green financial resource allocation in the automotive industry.

Keywords: green supply chain finance; feature selection; data augmentation; TabPFN; SHAP

1. Introduction

Against the background of the global transition toward sustainable development and the steady implementation of China's "dual carbon" goals, ESG has become an important benchmark for evaluating firms' long-term value creation and high-quality development capacity [1]. In the automotive industry, the overlap of green transformation, technological upgrading, and external uncertainty has made it necessary for firms to strengthen not only their green competitiveness but also supply chain resilience, green coordination, and life-cycle carbon reduction capabilities. Leading automotive manufacturers have increasingly embedded ESG principles into corporate strategy through supply chain localization and visualization, green supplier management, zero-carbon factories, low-carbon material procurement, power battery recycling, and circular economy practices. For supply chain finance, this transition has reshaped firms' operating models and risk sources, thereby imposing higher requirements on financial institutions in identifying SME credit risk, assessing supply chain stability, and evaluating long-term solvency [2]. Therefore, examining

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SME credit risk in the context of automotive green supply chain finance helps incorporate green supply chain factors into SME credit risk assessment and provides a more refined basis for understanding the relationships among green operational capability, supply chain collaboration, and credit risk.

Green supply chain finance is an emerging financing model formed by the integration of green finance and supply chain finance [3]. It emphasizes resource circulation and value co-creation across the industrial chain, and provides important financial support for the sustainable development of SMEs [4,5]. In China, green supply chain finance is still mainly reflected in the functional integration of green finance and supply chain finance at the operational level. Green finance refers to the allocation of financial resources to activities related to environmental improvement, climate change mitigation, and resource conservation, including investment, financing, and risk management services in environmental protection and clean energy sectors [6].

With the rapid expansion of new energy vehicles, the upgrading of auto components, and the greening of industrial chains, the automotive industry has evolved from a traditional manufacturing chain into a complex system characterized by technological intensity, data intensity, and environmental constraints. For SMEs in the automotive supply chain, which often face high R&D investment, long payback periods, and limited collateral capacity, these changes have increased their dependence on supply chain finance, trade finance, and data-driven credit evaluation [7]. In this context, the credit risk of green supply chain finance is no longer confined to debt repayment risk, but is jointly shaped by operational resilience, transaction authenticity, supply chain coordination, green compliance, and data traceability [8]. Meanwhile, the continuous innovation of supply chain finance has raised the demand for more accurate risk identification and management [9]. Thus, developing a scientific and comprehensive risk assessment framework for green supply chain finance is of both theoretical and practical significance.

Focusing on Chinese SMEs, this study makes four main contributions. First, using Chinese A-share-listed SMEs in the automotive industry as the research sample, this study develops an SME credit risk assessment framework in the context of automotive green supply chain finance that integrates the characteristics of green and low-carbon transformation with the risk formation mechanism of supply chain finance. Second, given the high dimensionality, weak correlations, and redundancy commonly observed in supply chain finance datasets, DE-LightGBM is introduced for feature selection to reduce noise and improve feature quality. Third, to address the class imbalance caused by the low proportion of risk-class SMEs, TabPFGen is employed for tabular data augmentation to improve sample representation and predictive performance. Fourth, TabPFN is applied for credit risk prediction and compared with traditional baseline models, while SHAP analysis is further used to identify key risk drivers and enhance model interpretability, providing decision support for commercial banks, investors, and policymakers.

The remainder of this paper is organized as follows. Section 2 reviews the literature on supply chain finance credit risk assessment, green supply chain finance risk identification, and related prediction models. Section 3 presents the proposed green supply chain finance credit risk evaluation system, the DE-LightGBM feature selection method, the TabPFGen augmentation strategy, and the TabPFN-SHAP prediction and interpretation framework. Section 4 reports the empirical results, including predictive performance, augmentation effects, robustness tests, and interpretability analysis. Section 5 presents practical recommendations, limitations, and future research directions, while Section 6 concludes the paper.

2. Literature Review

2.1. Theoretical Basis for Incorporating Green Supply Chain Factors

Regarding the formation mechanism of SME credit risk in supply chain finance scenarios, relevant foundational theories provide an important basis for understanding its sources and underlying mechanisms. Information asymmetry theory suggests that when creditors cannot fully observe the true quality of borrowing firms, adverse selection and moral hazard may increase the difficulty of credit risk identification [10]. Signaling theory further emphasizes that, under information incompleteness, observable information capable of conveying quality differences among economic agents can be used by external market participants to infer the potential quality of transaction partners [11]. Resource-based view argues that firms' competitive advantage derives from valuable, rare, inimitable, and non-substitutable resources and capabilities [12].

Existing studies on supply chain finance risk assessment have developed a relatively systematic theoretical foundation. Zhao and Li [13] developed an SME credit risk assessment indicator system for supply chain finance covering environmental, logistics, supply demand, information, and cooperation risks. Li et al. [14] showed that improving the management efficiency of supply chain enterprises can help reduce credit risk in supply chain finance. Huang and Sun [15] introduced a grey relational analysis model, emphasizing the importance of supply chain strength and real transaction backgrounds between upstream and downstream enterprises. Regarding the credit risk assessment of core enterprises, Huang et al. [16] identified suppliers' financial conditions and banks' risk control capabilities as key factors affecting supply chain finance risk in the telecommunications industry. Mou et al. [17] found that business performance and asset conditions are the two most important determinants of core enterprise credit risk, and emphasized the key role of core enterprises in credit enhancement within supply chain finance.

Green supply chain factors can reflect a firm's green transformation capability, supply chain coordination, and fulfillment of environmental responsibilities. Therefore, incorporating these factors into the credit risk assessment framework of automotive green supply chain finance is theoretically justified. Dai and Yang [18] found that digital empowerment promotes the green transformation of manufacturing firms by improving resource allocation efficiency, facilitating technological innovation, and generating green spillover effects along the industrial chain. Kou et al. [19] further indicated that corporate digitalization contributes to green transformation mainly by strengthening firms' innovation-driven capabilities. Liao and Zhang [20] demonstrated that green supply chain management plays a positive mediating role in the process through which carbon finance promotes corporate environmental innovation, suggesting that it can transform external green financial support into internal drivers of green innovation.

Taken together, these studies suggest that SME credit risk in green supply chain finance is shaped not only by financial conditions, but also by supply chain stability, core-enterprise support, digital capability, and green transformation practices.

2.2. Advanced Predictive Models for Supply Chain Finance Risk Assessment

Both academia and industry have actively explored the use of advanced technologies to improve the efficiency and accuracy of supply chain finance risk prediction. With their powerful data processing and analysis capabilities, machine learning technologies play a crucial role in risk assessment. Sang [21] constructed risk assessment models using genetic algorithms combined with SVM and BP neural networks, while Zhang et al. [22] adopted an SVM model optimized by the firefly algorithm. Zhu et al. [23] developed logistic regression, artificial neural network, and three types of two-stage hybrid models. Zhu et al. [24] further proposed an enhanced hybrid ensemble method, RS-MultiBoosting,

and empirically demonstrated its ability to improve predictive performance under limited supply chain finance risk data. Hou et al. [25] proposed an ISSA-LightGBM model for SME credit risk prediction in supply chain finance, where an improved sparrow search algorithm was used to optimize LightGBM hyperparameters, achieving better predictive performance and model interpretability.

Deep learning methods can automatically capture complex data representations and have achieved strong performance in prediction and classification tasks. Hu et al. [26] proposed a supply chain finance risk assessment model based on deep learning and fuzzy algorithms, in which a BP neural network was trained using particle swarm optimization. Xu et al. [27] applied a deep belief network to online supply chain finance risk assessment and showed that it outperformed SVM and logistic regression in both prediction accuracy and assessment rationality. Tang et al. [28] employed a deep convolutional neural network to improve risk identification in e-commerce supply chains, while Mojdehi et al. [29] proposed a hybrid TDA–GNN model for SCF credit risk assessment, demonstrating that the integration of topological and network-based features can improve SME credit risk prediction. Nalule et al. [30] developed a credit risk assessment model combining LSTM and CNN, and interpreted the model using explainable artificial intelligence methods such as LRP, SHAP, LIME, and integrated gradients. Zhang et al. [31] proposed DeepRisk, a multimodal deep learning model that integrates static enterprise demographic data and dynamic financing behavioral data for SME credit risk prediction in supply chain finance.

The Transformer architecture proposed by Vaswani et al. [32] represents a breakthrough in deep learning. With the growing application of deep learning to tabular data analysis, Hollmann et al. [33] proposed TabPFN, which has demonstrated strong performance in tabular classification tasks. Since its introduction, TabPFN has been increasingly applied across diverse domains due to its advantages in small-sample learning. Chen et al. [34] applied TabPFN to remote sensing data and improved the prediction accuracy of soil organic carbon content. Shang et al. [35] proposed an efficient battery thermal management method using TabPFN to predict complex thermal behavior and combined it with SHAP analysis to interpret the effects of features such as temperature and material properties on prediction outcomes. Li et al. [36] developed an Internet of Things network attack detection framework combining Lasso-RFE feature selection with an RF-TabPFN model, thereby improving both model performance and interpretability through a three-stage procedure. Xin et al. [37] proposed an interpretable TabPFN-based model for slag viscosity prediction, showing that TabPFN can achieve efficient and accurate prediction without extensive hyperparameter tuning. Recent advances in tabular foundation models have also provided new opportunities for synthetic data generation. Ma et al. [38] proposed TabPFGen, an energy-based tabular data generation framework built upon pretrained TabPFN, which can be used for data augmentation, class balancing, and imputation. Tugnoli et al. [39] further improved TabPFN-based synthetic data generation by incorporating causal structures to reduce spurious correlations and enhance structural fidelity. Therefore, this study introduces TabPFN into the credit risk prediction of automotive SMEs in green supply chain finance scenarios and integrates it with TabPFGen-based data augmentation to address the challenges of small sample size and class imbalance.

Based on the above review, the existing literature can be further extended in several respects. First, existing studies on supply chain finance credit risk have paid relatively limited attention to automotive green supply chain finance. Prior research on supply chain finance risk has mainly focused on traditional financial indicators, while the systematic integration of non-financial factors, such as green transformation, green supply chain management, digital capability, and supply chain operational coordination, remains relatively insufficient. Second, under the coexistence of small sample size, class imbalance,

and feature redundancy in supply chain finance credit risk data, how to achieve effective feature selection, minority-class sample augmentation, high-performance risk prediction, and interpretable analysis within a unified framework remains an important issue for further investigation. Third, green-related factors have often been used to explain green innovation, environmental performance, or green transformation outcomes, whereas their incremental predictive value for SME credit risk identification has not been fully examined.

To address these gaps, this study builds on existing risk assessment frameworks and constructs a multidimensional indicator system for credit risk prediction of automotive SMEs in green supply chain finance scenarios, with the aim of identifying more business-relevant non-financial risk drivers. Methodologically, DE-LightGBM is employed for feature selection to reduce redundancy and noise in high-dimensional heterogeneous data. TabPFGen-based data augmentation is then combined with TabPFN-based risk prediction to develop a prediction framework suitable for small-sample and imbalanced tabular data. To further examine the role of green-related factors, this study conducts an ablation experiment and a Wilcoxon signed-rank test to evaluate the incremental contribution of green-related variables to SME credit risk identification.

3. Materials and Methods

3.1. Dataset Description and Preparation

This study constructs a green supply chain finance credit risk evaluation system for SMEs in the automotive industry, as detailed in Appendix A Table A1. To ensure the systematicity and comprehensiveness of the indicator framework, the evaluation system was further refined and improved based on the existing literature [40–43], while taking into account the industry-specific characteristics of automotive supply chain finance and green-related factors. The framework adopts a hierarchical structure consisting of first-level, second-level, and third-level indicators. The first-level dimensions include macro-level indicators, comprehensive indicators of the financing enterprise, comprehensive supply chain indicators, and the credit support capacity of core enterprises. Specifically, the system integrates traditional financial indicators, such as solvency, profitability, operating efficiency, growth capacity, and firm size, with non-financial indicators related to supply chain cooperation, relationship intensity, collateral characteristics, trade credit and bargaining power, digitalization, green performance, and green supply chain performance. By incorporating green-related indicators such as ESG performance, greenhouse gas emissions, carbon information disclosure, green supply chain management, green production, and green products, the proposed framework provides a multidimensional basis for assessing SME credit risk in green supply chain finance.

In the context of automotive supply chain finance, the research sample consists of automotive SMEs listed on China's A-share market; core enterprises are typically large traditional automobile manufacturers with substantial fixed assets, strong industrial influence, and a central position within the supply chain. The sample period covers 2020–2024, and the data were obtained primarily from the Wind and CSMAR, as well as firms' publicly disclosed annual reports and related news reports. The dependent variable Y is interpreted as an indicator of SME credit risk in the context of automotive supply chain finance, capturing firm-level credit risk conditions relevant to credit assessment in supply chain finance scenarios. A firm-year observation was labeled as (Y = 1) if the enterprise experienced any publicly disclosed default-related event in that year, including overdue bill acceptance, overdue debt, or bond default, indicating a relatively high level of credit risk. Otherwise, it was labeled as (Y = 0), indicating non-default and a relatively low level of observed credit risk [31,44,45]. By integrating multiple supply chain records involving 210 SMEs, a dataset comprising 1050 observations and 56 credit risk assessment indicators was constructed,

including 112 risk samples and 938 non-risk samples. Given the limited sample size, the preprocessed dataset was randomly divided into a training set and a test set using an 8:2 train–test split to support robust model training and evaluation.

Prior to model training, data preprocessing was conducted to improve data quality and prevent information leakage. Missing values were imputed using the median values calculated from the training set. For selected financial variables with long-tailed distributions, such as total assets, operating revenue, total liabilities, and interest-bearing debt, logarithmic transformation was applied:

$$x_{ij}^{log} = \ln(x_{ij} + 1) \tag{1}$$

where x_{ij} denotes the original value of feature j for sample i. Categorical variables were converted into binary dummy variables through one-hot encoding. Finally, all numerical features were rescaled to the interval [0, 1] using Min–Max normalization:

$$x'_{ij} = \frac{x_{ij} - min(x_j)}{max(x_j) - min(x_j)} \tag{2}$$

where $min(x_j)$ and $max(x_j)$ denote the minimum and maximum values of feature j in the training set, respectively. All preprocessing parameters were estimated only from the training set and then applied to the testing set.

3.2. Feature Selection Model Based on DE-LightGBM

This study proposes a differential-evolution-optimized LightGBM feature selection framework, denoted as DE-LightGBM. The method combines the global search capability of differential evolution with the nonlinear modeling and feature evaluation advantages of LightGBM, aiming to identify an informative feature subset from high-dimensional and redundant supply chain finance credit risk data.

Let the original feature set be $X = \{x_1, x_2, \ldots, x_d\}$. Each individual in the differential evolution population is encoded as a continuous vector:

$$z_i = (z_{i1}, z_{i2}, \ldots, z_{id}), \quad z_{ij} \in [0, 1] \tag{3}$$

The continuous vector is then converted into a binary feature mask according to a predefined threshold θ:

$$m_{ij} = \begin{cases} 1 & z_{ij} \geq \theta \\ 0 & z_{ij} < \theta \end{cases} \tag{4}$$

where $m_{ij} = 1$ indicates that the corresponding feature is selected. For each selected feature subset, a LightGBM classifier is trained and evaluated through cross-validation. Considering the class imbalance in SME credit risk identification, the risk class F1-score of the risk class is used as the fitness function:

$$F(S_i) = \frac{1}{K} \sum_{k=1}^{K} F1_{risk}^{(k)}(S_i) \tag{5}$$

where K denotes the number of cross-validation folds. During the optimization process, differential mutation, crossover, and selection are iteratively performed to update candidate feature subsets. For a target individual, the mutation operation is defined as:

$$v_i = z_{r1} + F(z_{r2} - z_{r3}) \tag{6}$$

where F is the mutation factor, and z_{r1}, z_{r2}, and z_{r3} are three distinct individuals randomly selected from the population. If the trial individual obtains a fitness value no lower than that of the current individual, it replaces the current individual.

Through this iterative search process, DE-LightGBM removes redundant and weakly informative variables while retaining key risk-related features. The selected feature subset is then used as the input for TabPFGen-based data augmentation and the TabPFN credit risk prediction model. The detailed procedure is illustrated in Figure 1.

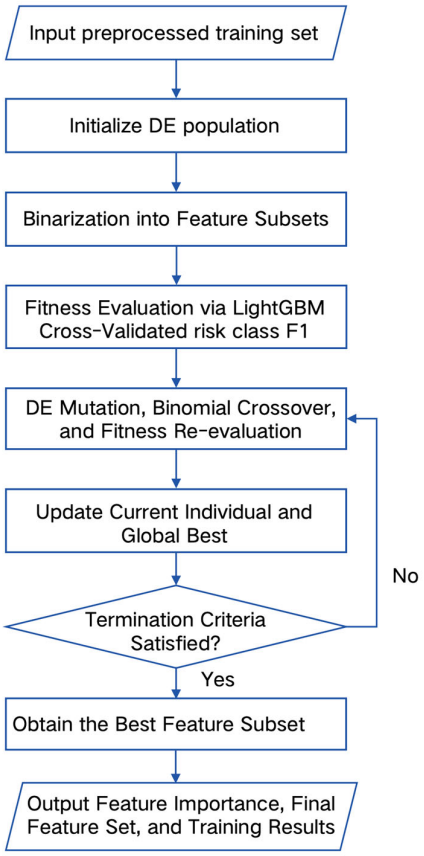


Figure 1. Flowchart of the DE-LightGBM Feature Selection Framework.

3.3. Data Augmentation Based on TabPFGen

To address the class imbalance problem in automotive-industry SME credit risk data in the context of supply chain finance, this study adopts TabPFGen to generate synthetic samples for the minority risk class. Since all features were normalized to the interval [0, 1], the generated samples were clipped to the same range to ensure numerical consistency with the original training data. For one-hot encoded categorical variables, a post-processing correction was applied by assigning 1 only to the category with the largest generated value within each dummy-variable group and setting the remaining categories to 0, thereby preserving categorical exclusiveness and business logic.

To improve the business validity and reproducibility of the synthetic samples, this study introduced a set of post-generation business-rule constraints before model training. These constraints were constructed using only the training set and were mainly applied to features with clear categorical exclusiveness, binary business meanings, or explicit joint-relationship structures. Specifically, categorical variables were corrected for one-hot consistency, binary green-practice variables were mapped to admissible values, and macro-industry and debt-structure variables were projected to observed training-set combinations

where applicable. The detailed constraint types, involved features, and post-processing rules are summarized in Appendix A Table A2.

In addition, the nearest neighbor distance ratio (NNDR) was used to filter synthetic samples and reduce the risk of near duplication:

$$\mathrm{NNDR}_i = \frac{\mathrm{d_{i,1}}}{d_{i,2}} \tag{7}$$

where $d_{i,1}$ and $d_{i,2}$ denote the distances from the i-th synthetic sample to its nearest and second-nearest original risk-class samples, respectively. Synthetic samples were retained only when their NNDR values fell within the interval [0.01, 0.95]. This interval was used as a permissive quality-control criterion rather than as a performance-tuning hyperparameter. The lower bound of 0.01 was introduced to remove only extreme near-duplicate cases in which a synthetic sample is disproportionately close to a single real risk-class observation. A substantially higher lower bound may excessively remove synthetic samples located close to the real minority-class manifold, thereby reducing distributional fidelity and discarding potentially informative samples near the minority-class decision boundary. The upper bound of 0.95 was used to exclude extreme cases in which a synthetic sample has almost the same distance to its nearest and second-nearest real risk-class observations, indicating ambiguous local nearest-neighbor support. Since this upper bound is close to 1, it removes only highly ambiguous samples while retaining most synthetic observations that may help expand the minority-class feature space.

To further assess whether the synthetic samples improve generalization rather than merely reproducing local patterns of the original risk-class observations, the utility of TabPFGen-based augmentation was evaluated on real, non-augmented validation data. Specifically, in the cross-validation procedure, synthetic samples were generated and added only to the corresponding training fold, while the validation fold was kept unchanged and consisted exclusively of real observations. Under this design, any performance improvement after augmentation is evaluated on unseen real samples rather than on synthetic samples, thereby providing a more appropriate assessment of the generalization benefit of the generated minority-class observations.

3.4. Risk Assessment Model Based on TabPFN-SHAP

TabPFN is a Transformer-based Prior-Data Fitted Network for tabular prediction. It leverages a learned tabular data prior and in-context learning to perform efficient classification in small-sample settings, thereby reducing the need for extensive task-specific model training. This property makes it suitable for SME credit risk prediction in green supply chain finance scenarios, where the dataset is small, heterogeneous, and imbalanced, and where nonlinear interactions may exist among financial, supply chain, green-related, and digital indicators. Compared with deep learning models trained from scratch, which typically require larger sample sizes and more extensive hyperparameter tuning, TabPFN reduces the dependence on large task-specific training data. In addition, TabPFGen-based augmentation is methodologically compatible with TabPFN prediction: TabPFGen enriches the representation of scarce risk-class observations, while TabPFN uses the augmented tabular context to improve minority-class risk identification. Therefore, this study further combines TabPFN with feature selection, TabPFGen-based data augmentation, and constrained threshold optimization to improve the recognition of high-risk enterprises.

This study employs TabPFN-v2 as the core credit risk prediction model and compares its performance with several classical machine learning benchmarks. Specifically, five baseline models are considered, including XGBoost, CatBoost, LightGBM, logistic regression, and random forest. Bayesian optimization was applied to the tree-based boosting models,

including XGBoost, CatBoost, and LightGBM, to obtain more suitable hyperparameter configurations. Ten-fold cross-validation is adopted during model training to obtain more robust performance estimates. In addition, this study adopts a constrained dynamic threshold optimization strategy, in which the threshold yielding the highest F1-score is selected as the final decision threshold from the candidate thresholds that ensure a risk-class recall of no less than 80% during internal cross-validation on the training set.

For model interpretability, the SHAP method is further introduced to explain the prediction results of TabPFN. Compared with traditional feature importance measures, SHAP not only evaluates the overall importance of features but also illustrates how changes in feature values affect model outputs positively or negatively. SHAP values for the TabPFN model are calculated using the get_shap_values function provided by the tabpfn_extensions library. Based on the calculated SHAP values, SHAP summary plots and bar plots are further generated to identify the key factors influencing SME credit risk prediction in supply chain finance.

4. Results

4.1. Iteration Sensitivity Analysis of Feature Selection

To examine the effect of the number of iterations on predictive performance and feature selection stability, an iteration sensitivity analysis was further conducted. No early stopping strategy was applied during the sensitivity analysis stage to ensure a fair comparison of model performance under different iteration settings. In the formal feature selection stage, the optimal number of iterations was used as the maximum iteration limit, and an early stopping mechanism was then introduced to reduce redundant searches and improve training efficiency. As shown in Table 1, 50 iterations achieved a relatively high F1_score with a low standard deviation in the number of selected features, indicating a favorable balance between predictive performance and feature selection stability. Although 60 iterations yielded the highest F1 score of 0.71, its Jaccard index decreased to 0.42, suggesting that the consistency of the selected feature subsets declined while classification performance improved. The moderate level of the Jaccard index also reflects the multi-source heterogeneity and variable substitutability inherent in supply chain finance risk formation. Therefore, the maximum number of iterations was set to 60 in the formal feature selection process. To improve computational efficiency and avoid redundant searches, an early stopping criterion was introduced, whereby the optimization process was terminated if no improvement in the fitness value was observed for 10 consecutive iterations.

Table 1. Iteration Sensitivity Analysis of DE-LightGBM Feature Selection.

Max_Iter	F1 (Y = 1)	Std_F1	n_Features	Std_n_Features	Jaccard
10	0.6332	0.0194	35	3.4205	0.4699
20	0.6648	0.0114	29	5.0200	0.4274
30	0.6773	0.0072	30	2.5884	0.4554
40	0.6826	0.0122	31	3.5777	0.4145
50	0.7039	0.0126	28	1.5811	0.4693
60	0.7099	0.0178	31	1.3416	0.4229
70	0.7063	0.0141	27	1.9494	0.4578

4.2. TabPFGen Sample Augmentation Evaluation

4.2.1. Sensitivity Analysis of TabPFGen Augmentation Size

To further examine the effect of minority-class augmentation size on predictive performance and generalization stability, this study conducts a sensitivity analysis of the number of TabPFGen-generated samples based on five-fold cross-validation, as reported in Table 2. The "total number of augmented samples" refers to the cumulative number of synthetic samples generated across the training folds in the five-fold cross-validation procedure, while the validation folds remain unaugmented to avoid data leakage.

Table 2. Sensitivity Analysis Results of TabPFGen Augmentation Size.

Total Number	F1 Score (Y = 1)	Std F1 Score	Precision (Y = 1)	Recall (Y = 1)	AUC	AUC-PR
50	0.8707	0.0408	0.8452	0.9000	0.9842	0.9097
100	0.8478	0.0653	0.8120	0.8889	0.9815	0.8875
150	0.8722	0.0507	0.8376	0.9111	0.9841	0.9048
200	0.8627	0.0597	0.8029	0.9333	0.9808	0.8901
250	0.8699	0.0500	0.8686	0.8778	0.9833	0.9066
300	0.8495	0.1112	0.8141	0.9111	0.9811	0.8956
400	0.8632	0.0632	0.8497	0.8889	0.9810	0.8955

As shown in Table 2, the number of augmented samples has a notable effect on the risk identification performance of the TabPFN model; however, a larger augmentation size does not necessarily lead to better performance. When the total number of augmented samples is 150, the model achieves the highest mean F1 score of 0.87, with the recall for the risk class reaching 0.91. When the augmentation size increases to 200, the recall for the risk class further improves to 0.93, but the precision decreases to 0.80, indicating that the model becomes more sensitive to risk-class samples while also increasing the likelihood of misclassifying non-risk firms as risky firms. When the total number of augmented samples reaches 300 or more, further augmentation does not produce stable marginal performance gains.

Overall, moderate data augmentation can alleviate the majority-class bias caused by the scarcity of risk-class samples. The sensitivity analysis showed that retaining 150 synthetic risk-class samples achieved a favorable balance between predictive performance and model stability. Since the full training set used for final model fitting was approximately 1.25 times the data scale adopted in the training-set-only cross-validation experiment, the augmentation size was adjusted proportionally for the final training stage. Accordingly, 200 synthetic risk-class samples that passed the business-rule constraints and NNDR filtering were retained and merged with the original training set in the formal training stage.

4.2.2. Visual Assessment of TabPFGen Data Augmentation

1. Changes in Category Distribution and NNDR Test

After minority-class augmentation using TabPFGen, the class imbalance problem in the training set was substantially alleviated. As shown in Figure 2, the number of non-risk samples remained unchanged at 750, whereas the number of risk samples increased from 90 to 290. The 200 additional samples were synthetic risk-class observations retained after applying business-rule constraints and NNDR-based filtering. Consequently, the imbalance ratio between non-risk and risk samples in the training set decreased from approximately 8.33:1 before augmentation to 2.59:1 after augmentation. This ratio is relatively acceptable for subsequent prediction using the TabPFN model, which, as a prior-data-fitted Transformer-based model, is well suited to learning from limited and

imbalanced tabular data. Figure 3 presents the NNDR distribution of the retained synthetic samples. The NNDR values of the retained samples are mainly distributed within the range of 0.01 to 0.09, and all samples satisfy the predefined NNDR filtering threshold of 0.01. This indicates that synthetic samples excessively close to the original risk-class observations were effectively removed. Therefore, the retained synthetic samples are not simple replications of the original risk samples; rather, they introduce a certain degree of local variation while preserving the fundamental distributional characteristics of the minority class.

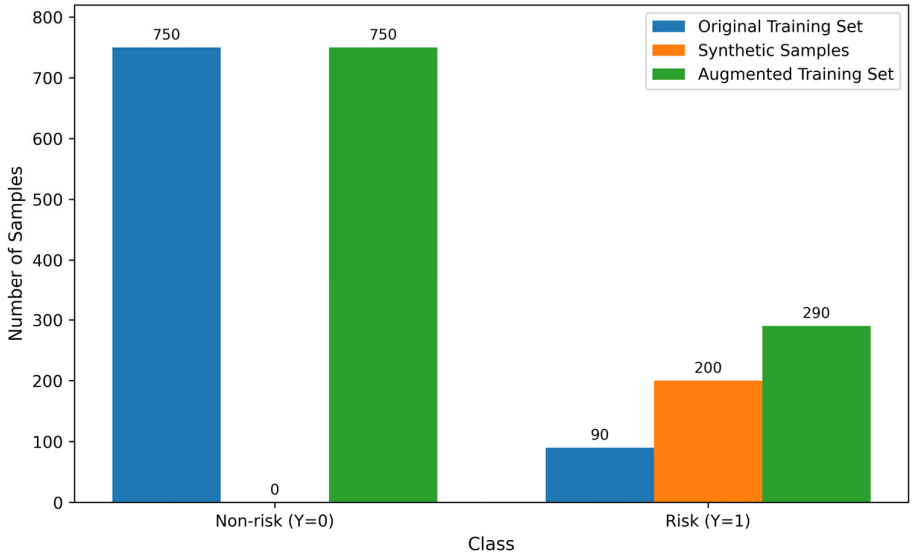


Figure 2. Class Distribution Before and After TabPFGen Augmentation.

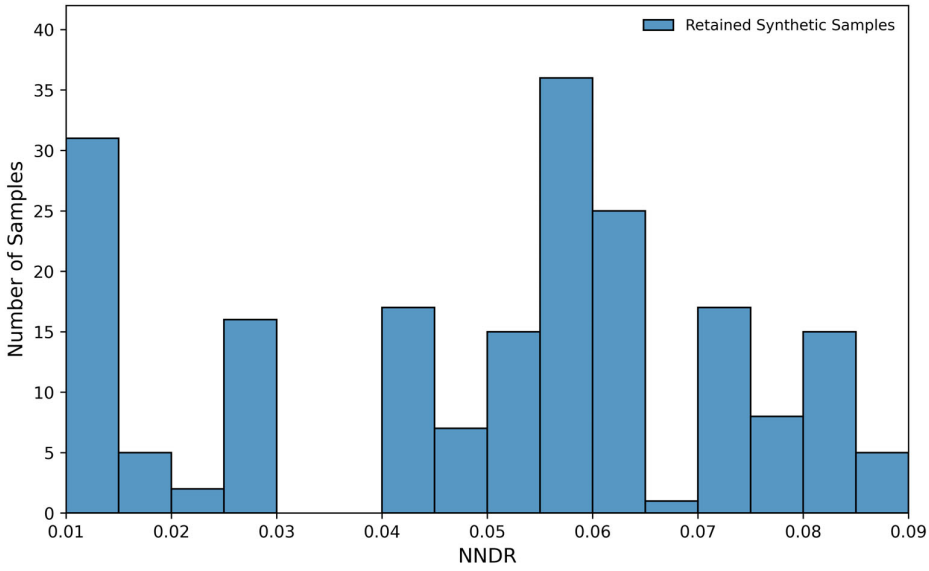


Figure 3. NNDR Distribution of TabPFGen-Generated Synthetic Samples.

2. Generated Sample Visualization Analysis

Figure 4 presents the t-SNE visualization of the original and TabPFGen-generated risk-class samples. The synthetic samples are generally distributed around the original risk samples and exhibit moderate local expansion, without forming clearly isolated clusters. Although t-SNE is only a qualitative visualization method and cannot provide formal statistical evidence of distributional equivalence, the absence of distinct anomalous clusters indicates that the TabPFGen-generated samples do not substantially deviate from the

original risk-class samples. The augmented samples were subsequently used for predictive model training to improve the model's learning of risk-class patterns.

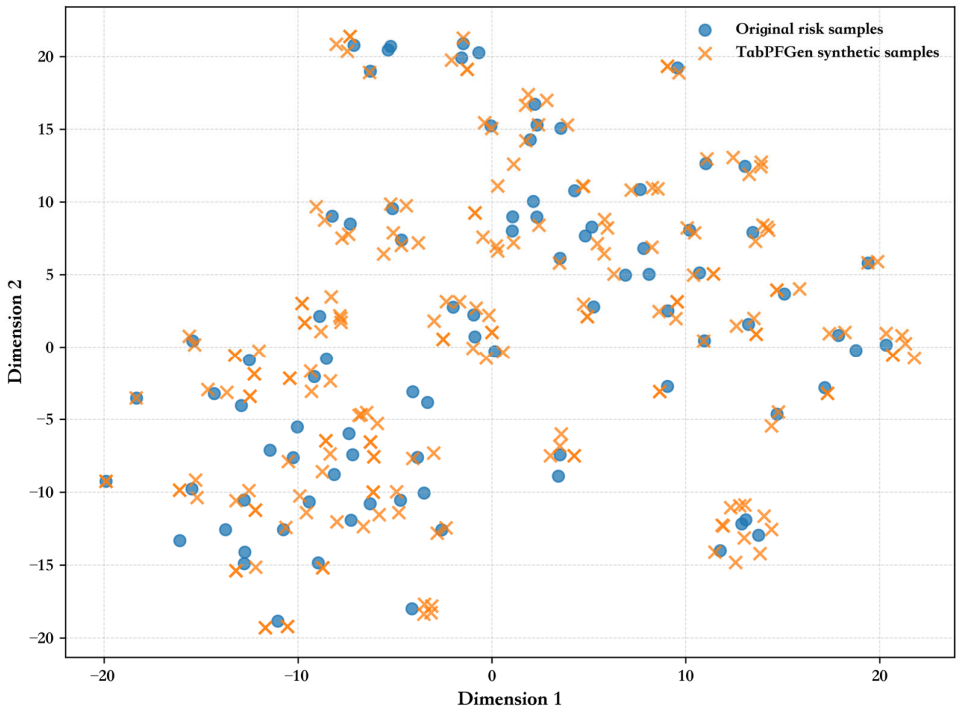


Figure 4. t-SNE Visualization of Original and Synthetic Risk-class Samples.

3. Statistical Consistency Test

In addition to the visual assessment, a statistical consistency test was conducted to further examine whether the TabPFGen-generated samples preserved the main distributional characteristics of the original risk-class samples. The results show that the mean KSComplement for continuous variables is 0.9117, indicating a high degree of similarity in marginal distributions. The mean TVComplement for discrete variables is 0.9325, suggesting that the marginal distributions of discrete features are well retained. Moreover, the CorrelationSimilarity reaches 0.9646, indicating that the inter-variable correlation structure is largely preserved, while the mean ContingencySimilarity is 0.8866, showing that the joint distribution patterns of discrete variables are reasonably maintained. Overall, these results suggest that the generated risk-class samples exhibit good statistical consistency with the original minority-class observations. Detailed variable-level results are reported in Appendix A Tables A3 and A4.

4.3. Models' Performance on the Test Set

To evaluate the predictive performance of TabPFN and the baseline models, three datasets were constructed: Raw, the preprocessed original dataset; Selected, the dataset obtained after DE-LightGBM feature selection; Augmented, the data augmentation by applying TabPFGen after feature selection. The comparison across these datasets allows us to examine the effects of feature selection and data augmentation on model performance. Model performance was evaluated using Accuracy, F1-score, AUC, Precision, and Recall, covering both overall classification performance and risk-class identification ability. For the independent test-set evaluation, model selection and threshold optimization were conducted using only the training set, and the final model was then refitted on the full training set and evaluated on the unchanged independent test set.

To ensure fairness and consistency in model comparison, all models were evaluated on the same test set, and the results are reported in Table 3. Overall, TabPFN outperformed the baseline models, including XGBoost, CatBoost, LightGBM, LR, and RandomForest, across all three datasets. On the augmented dataset, TabPFN achieved the highest point-estimate performance among the compared models under the current test-set setting, with an overall accuracy of 97.62% and a risk-class F1-score of 88.37%. Its Precision and Recall for the risk class reached 0.9048 and 0.8636 respectively, suggesting that the model showed a relatively strong ability to identify minority risk enterprises under the current test-set setting.

Table 3. Comparative Performance of TabPFN and Baseline Models on the Independent Test Set.

Model	Training Set	Accuracy	F1 Score (Y = 1)	AUC	Precision (Y = 1)	Recall (Y = 1)	Precision (Y = 0)	Recall (Y = 0)
XGBoost	Raw	0.9476	0.7556	0.9352	0.7391	0.7727	0.9733	0.9681
CatBoost	Raw	0.9476	0.7660	0.9399	0.7200	0.8182	0.9784	0.9628
LightGBM	Raw	0.9571	0.7907	0.9459	0.8095	0.7727	0.9735	0.9787
LR	Raw	0.8857	0.5862	0.8776	0.4722	0.7727	0.9713	0.8989
RF	Raw	0.9333	0.7200	0.9252	0.6429	0.8182	0.9780	0.9468
TabPFN	Raw	0.9667	0.8373	0.9819	0.8571	0.8182	0.9788	0.9840
XGBoost	Selected	0.9524	0.7826	0.9375	0.7500	0.8182	0.9785	0.9681
CatBoost	Selected	0.9571	0.8000	0.9526	0.7826	0.8182	0.9786	0.9734
LightGBM	Selected	0.9524	0.7917	0.9385	0.7308	0.8636	0.9837	0.9628
LR	Selected	0.8810	0.5902	0.8616	0.4615	0.8182	0.9766	0.8883
RF	Selected	0.9476	0.7660	0.9421	0.7200	0.8182	0.9784	0.9628
TabPFN	Selected	0.9714	0.8571	0.9877	0.9000	0.8182	0.9789	0.9894
XGBoost	Augmented	0.9476	0.7755	0.9350	0.7037	0.8636	0.9836	0.9574
CatBoost	Augmented	0.9619	0.8095	0.9570	0.8500	0.7727	0.9737	0.9840
LightGBM	Augmented	0.9524	0.7826	0.9399	0.7500	0.8182	0.9785	0.9681
LR	Augmented	0.8714	0.5714	0.8664	0.4390	0.8182	0.9763	0.8777
RF	Augmented	0.9476	0.7660	0.9376	0.7200	0.8182	0.9784	0.9628
TabPFN	Augmented	0.9762	0.8837	0.9886	0.9048	0.8636	0.9841	0.9894

Regarding the effects of different data processing strategies, feature selection generally improved model performance. For example, both TabPFN and CatBoost showed improvements in Accuracy, F1-score, and AUC after feature selection, suggesting that DE-LightGBM can effectively eliminate redundant features and noisy variables, thereby enhancing the model's ability to capture key risk-related information.

Further comparison of the augmented dataset indicates that the TabPFGen-driven minority-class augmentation strategy was more beneficial to TabPFN, whereas its improvements for conventional machine learning models were relatively limited. This result may be attributed to the compatibility between TabPFGen and TabPFN in small-sample tabular learning. The synthetic risk-class observations produced by TabPFGen expand the coverage of minority samples in the feature space and provide richer contextual information for TabPFN to infer class boundaries. By contrast, conventional tree-based models already have strong capabilities in nonlinear fitting and feature interaction learning, especially after feature selection has removed redundant information. Therefore, the additional synthetic observations may provide only limited marginal information for these models. Moreover, the augmented risk-class observations mainly complement the local distribution of minority samples; although this may improve model sensitivity to the risk class, it may also alter local sample density and lead to a trade-off between Precision and Recall.

Figure 5 presents the confusion matrix of the TabPFN model trained on the augmented dataset and evaluated on the unchanged independent test set. The model correctly identified 19 of the 22 risk-class observations, while three risky observations were misclassified

as non-risk. In addition, only two non-risk observations were incorrectly classified as risky, corresponding to a false negative rate of 13.64% for risky firms and a false positive rate of 1.06% for non-risk firms. These results suggest favorable point-estimate performance of TabPFN trained on the augmented dataset, particularly in terms of risk detection and false-alarm control.

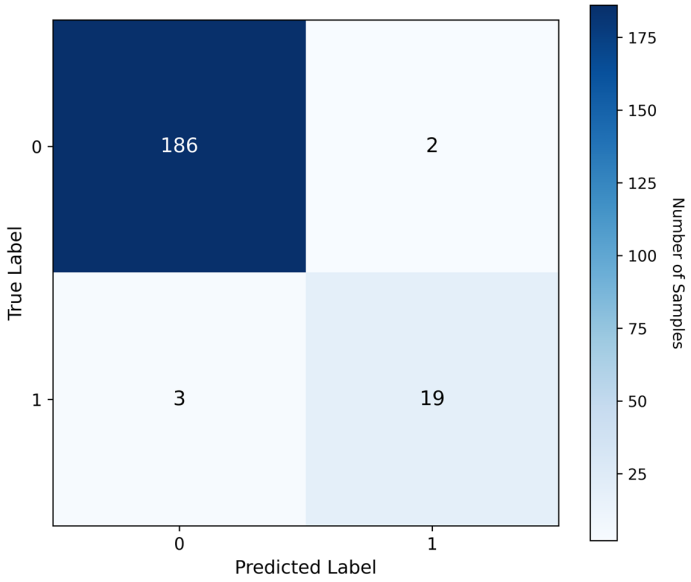


Figure 5. Test Set Confusion Matrix of TabPFN Trained on the Augmented Dataset.

Given the limited number of risk-class observations in the independent test set, threshold-dependent minority-class metrics may be sensitive to a small number of classification errors. Therefore, we further report 95% confidence intervals for the main test-set metrics in Appendix A Table A5 to quantify the uncertainty of these estimates. Compared with TabPFN trained on the selected dataset, TabPFN trained on the augmented dataset yielded higher point estimates for risk-class F1-score, Recall, AUC, and AUC-PR, suggesting that TabPFGen-based minority-class augmentation may improve the model's sensitivity to risky firms. However, the relatively wide confidence intervals for Precision, Recall, and F1-score indicate that the minority-class metrics remain subject to non-negligible uncertainty, although the model retains favorable point-estimate performance under the current test-set setting. Overall, these results provide additional support for the potential value of TabPFN trained on the augmented dataset in SME credit risk identification in the context of supply chain finance under the current test-set setting.

In addition to the independent test set evaluation, ten-fold cross-validation was further conducted on the training set to examine the statistical stability of TabPFN under different data processing strategies. In each fold, the validation fold was kept unchanged, while TabPFGen-based augmentation was applied only to the corresponding training fold. The risk class F1-scores obtained from the ten folds were then compared using the Wilcoxon signed-rank test, as reported in Table 4. TabPFN trained on the Augmented dataset obtains a higher mean F1-score (0.9064) than other datasets, with a relatively lower standard deviation (0.0804). The paired Wilcoxon signed-rank test indicates that the differences are statistically significant for both comparisons, namely Augmented versus Selected ($p = 0.008$) and Augmented versus Raw ($p < 0.001$). The corresponding 95% confidence intervals exclude zero, suggesting that the observed improvements are relatively consistent across the cross-validation folds. Overall, these results suggest that feature selection helps reduce redundant and noisy variables, while data augmentation further mitigates the scarcity of risk-class samples, thereby enhancing both predictive performance and model stability.

Table 4. Wilcoxon Signed-Rank Test Results for the TabPFN across Different Datasets.

Dataset	F1 Score (Y = 1)	SD	Compared To	*p*-Value	Mean Diff	95% CI
Augmented	0.9064	0.0804	-	-	-	-
Selected	0.8215	0.1524	Augmented	0.008	0.0849	[0.0111, 0.1234]
Raw	0.7950	0.1535	Augmented	<0.001	0.1114	[0.0653, 0.1451]

4.4. SHAP Analysis

To further explain how the model identifies SME credit risk in the context of automotive supply chain finance, this study applies SHAP analysis to interpret the prediction results. In Figure 6, Panel A illustrates the direction and distribution of feature effects on the model output, while Panel B reports the mean absolute SHAP values for risk and non-risk samples, thereby indicating the relative importance of each feature.

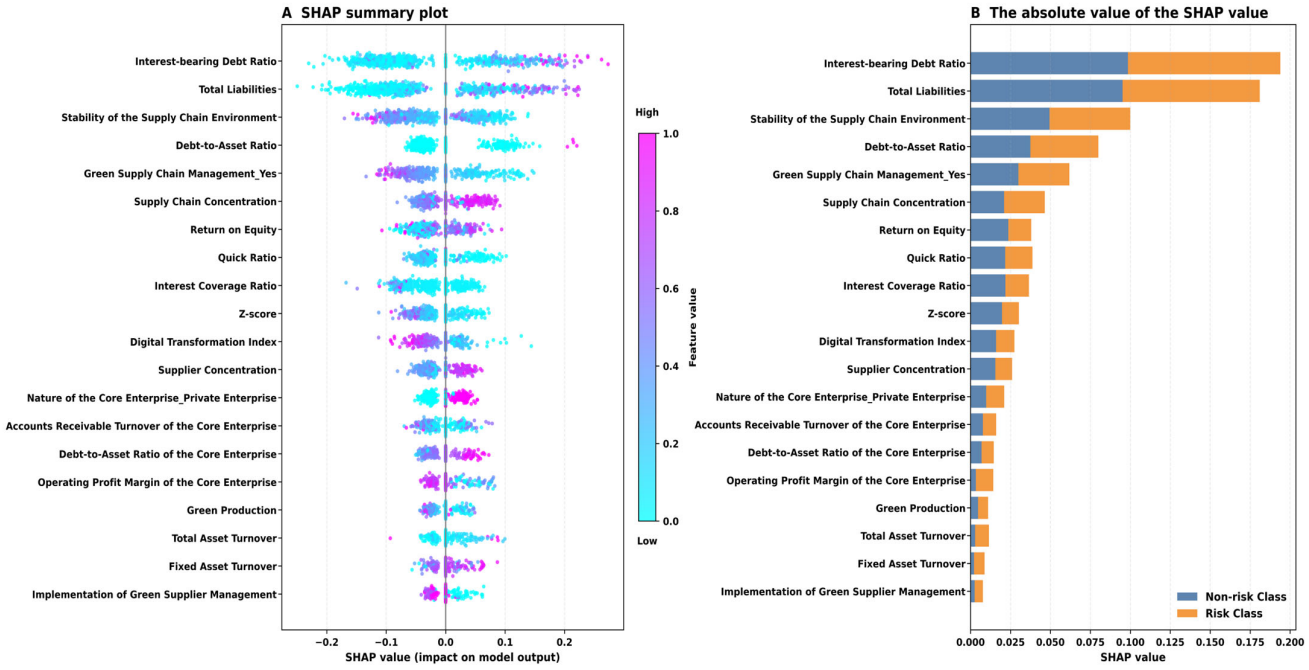


Figure 6. SHAP Summary and Feature Importance Analysis of the TabPFN Model.

Overall, the model primarily relies on debt pressure, supply chain stability, solvency, core enterprise characteristics, green supply chain management, and digitalization level. Among these factors, the interest-bearing debt ratio and total liabilities exhibit the largest explanatory contributions, suggesting that debt burden remains a key determinant of SME credit risk. A higher proportion of interest-bearing debt increases rigid repayment pressure and liquidity constraints, especially when operating cash flow is insufficient or financing conditions tighten. Similarly, larger total liabilities weaken financial flexibility and increase firms' vulnerability to operational fluctuations and external shocks.

Supply chain-related variables also play an important role. The stability of the supply chain environment ranks highly, indicating that a stable supply chain can reduce operational uncertainty and credit risk. Supply chain concentration and supplier concentration further reflect the potential transmission of risk through highly dependent trading relationships. From the perspective of signaling theory, variables such as supply chain stability, relationship intensity, supply chain concentration, and core-enterprise characteristics provide observable signals for financial institutions to assess the quality of firms' trading relationships, the degree of repayment assurance, and the strength of external credit sup-

port. Higher supply chain concentration and supplier concentration may also indicate firms' dependence on a limited number of trading partners, thereby increasing the possibility of risk transmission along the supply chain.

The relatively high ranking of green supply chain management indicates that green-related features provide incremental explanatory value for risk identification. Firms with green supply chain management practices tend to be associated with a lower predicted probability of risk, indicating that this variable serves as an important non-financial signal rather than merely an environmental indicator. Based on information asymmetry theory and signaling theory, variables such as Green Supply Chain Management, Carbon Information Disclosure, and Green Transformation Disclosure can enhance the observability of firms' green operations and governance quality. Firms with such practices tend to exhibit stronger compliance awareness, higher information transparency, and better coordination with upstream and downstream partners.

The digital transformation index also shows certain explanatory relevance. From the perspectives of the resource-based view and dynamic capabilities theory, digitalization-related indicators may reflect firms' ability to improve operational transparency, strengthen supply chain embeddedness, integrate internal and external resources, and adapt to technological change to some extent. Higher levels of digitalization can further enhance data traceability and information processing, helping financial institutions verify transaction records, inventory information, accounts receivable, and operating conditions more effectively.

In summary, the SHAP interpretation results show that the model identifies supply chain finance credit risk among SMEs primarily through factors related to debt pressure, solvency, supply chain stability, core enterprise quality, green supply chain management, and digitalization. Among these factors, green supply chain management and digital transformation provide important non-financial signals, as they may reflect firms' governance capability, environmental compliance, information transparency, and supply chain coordination. These characteristics help the model distinguish risky enterprises from non-risky ones more effectively. It should be noted that SHAP values provide post-hoc explanations of the fitted model and should not be interpreted as evidence of causal effects.

4.5. Incremental Value of Green-Related Features

The green-related features in this study comprise two second-level indicators. The first is Green Performance under Comprehensive Indicators of the Financing Enterprise, which includes three third-level indicators: Wind ESG Rating, Wind ESG Composite Score, and Greenhouse Gas Emissions. The second is Green Supply Chain Performance under Comprehensive Supply Chain Indicators, which includes seven third-level indicators: Carbon Information Disclosure, Green Supply Chain Management, Green Transformation Disclosure, Green Supply Chain Management Strategy, Implementation of Green Supplier Management, Green Production, and Green Products.

To examine whether green-related features provide additional predictive information for supply chain finance credit risk identification, this study conducts an ablation experiment by comparing two feature sets: one excluding all green-related features and the other retaining the complete feature set. To ensure comparability, the same experimental protocol was independently applied to both the full feature set and the feature set excluding green-related variables, including feature selection, data augmentation, TabPFN model training, and threshold optimization. As shown in Table 5, compared with the feature set excluding green-related variables, the complete feature set improves Accuracy from 0.9571 to 0.9762, the risk-class F1-score from 0.8163 to 0.8837, AUC from 0.9778 to 0.9886, and risk-class Precision from 0.7407 to 0.9048. These results indicate that green-related features provide useful incremental information for improving overall discriminative performance and the

reliability of positive risk predictions. However, the inclusion of green-related features also leads to a decrease in risk-class Recall from 0.9091 to 0.8636. This indicates that green-related features improve the model's overall performance, risk-class Precision, risk-class F1-score, and AUC, while slightly reducing sensitivity to some risky firms. The threshold optimization procedure in this study already imposed a risk-class Recall constraint during internal cross-validation on the training set. Nevertheless, in practical credit risk assessment, the final threshold choice should still be considered together with institutional risk preferences.

Table 5. Performance Comparison of TabPFN with and without Green-related Features.

Feature Set	Accuracy	F1 Score (Y = 1)	AUC	Precision (Y = 1)	Recall (Y = 1)	Precision (Y = 0)	Recall (Y = 0)
All Features	0.9762	0.8837	0.9886	0.9048	0.8636	0.9841	0.9894
Exclude Green-related Features	0.9571	0.8163	0.9778	0.7407	0.9091	0.9891	0.9628

Given the theoretically motivated directional hypothesis that green-related features improve credit risk prediction, a one-sided paired Wilcoxon signed-rank test was used. Wilcoxon signed-rank test further supports the incremental predictive value of green-related features in supply chain finance credit risk identification. As reported in Table 6, the model using all features achieved a higher mean F1 score than the model excluding green-related features. This improvement is statistically significant under the one-sided Wilcoxon signed-rank test, which assumes that the inclusion of green-related features improves predictive performance ($p = 0.0391$).

Table 6. Wilcoxon Signed-Rank Test for the Incremental Predictive Value of Green-Related Features.

Feature Set	F1 Score (Y = 1)	SD	*p*-Value	Mean Diff	95% CI
All Features	0.9064	0.0804	-	-	-
Exclude Green-related Features	0.8015	0.1115	0.0391	0.1049	[0.0048, 0.2050]

From an economic and risk-management perspective, green-related features provide non-financial information that extends beyond traditional financial indicators. Green performance and green supply chain practices reflect firms' environmental compliance, governance quality, information transparency, and supply chain coordination. These factors can help reduce regulatory risk and information asymmetry, thereby providing financial institutions with valuable signals for assessing the operational stability and credit risk exposure of SMEs.

Overall, the ablation results demonstrate that green-related features provide significant incremental predictive value for identifying supply chain finance credit risk among automotive SMEs. The improvement in both predictive performance and cross-validation stability indicates that green factors can effectively complement financial, operational, and supply chain variables, thereby enhancing the robustness and explanatory power of the credit risk prediction model. Therefore, green-related features should not be regarded merely as supplementary environmental variables; rather, they contain meaningful risk-identification information. This finding supports the necessity of incorporating green-related factors into supply chain finance credit risk assessment, while their practical use should be combined with threshold adjustment and risk-preference considerations.

5. Discussion

The empirical results indicate that green-related features provide significant incremental predictive value for supply chain finance credit risk identification. This conclusion is further supported by the SHAP analysis, which shows that green practice variables, including Green Supply Chain Management, Implementation of Green Supplier Management, and Green Production, make substantial contributions to the model's predictions. These variables are not merely supplementary non-financial indicators; rather, they serve as informative risk signals that capture firms' environmental compliance, governance quality, supply chain coordination, and operational sustainability, thereby helping explain the observed improvement in overall discriminative performance. From a practical perspective, financial institutions should incorporate green governance performance, carbon disclosure, green supply chain management practices, and digital capability into credit access evaluation, credit line pricing, and early-warning monitoring. Core enterprises should further strengthen transaction data sharing, accounts receivable confirmation, and green collaborative governance across the supply chain. Meanwhile, regulators should promote standardized disclosure of green and supply chain data to reduce information asymmetry and improve the risk-pricing capability of green supply chain finance.

Despite these findings, this study still has several limitations, and future research can be improved in the following three aspects. First, the dependent variable in this study is constructed from publicly disclosed firm-level credit risk events, including overdue bill acceptance, overdue debt, and bond default. Although these events are relevant to credit assessment in supply chain finance scenarios, they do not necessarily represent defaults on specific supply chain finance transactions. Therefore, future research could collect transaction-level supply chain finance data to validate the proposed framework at a more granular business level. In addition to the existing financial indicators and supply chain relationship indicators, subsequent research may seek cooperation with enterprises and further integrate multi-source heterogeneous data, such as corporate financial data, transaction records, purchase orders and contracts, logistics fulfillment information, warehouse supervision records, tax invoices, judicial litigation records, public opinion information, and industrial chain network relationships. This would help overcome the limitation of relying solely on publicly available annual data for static risk identification. Future research may also incorporate sample data covering a wider range of industries and enterprise types to enhance the generalizability of the model conclusions.

Second, at the methodological level, future studies can further strengthen the dynamic modeling capability of risk prediction models. To better capture the accumulation and transmission of risk over time, future research could incorporate time-series deep learning models and online learning mechanisms, allowing the model to update incrementally as new data become available and thereby improving its predictive performance in complex and evolving environments. Future studies may also introduce artificial intelligence large models and graph neural networks into supply chain finance risk prediction. At the level of data trustworthiness and physical sensing, blockchain and Internet of Things technologies can be integrated to improve the authenticity and traceability of underlying supply chain finance data. At the level of advanced applications, future research may explore the use of digital twins and federated learning in supply chain finance risk management. Federated learning can further enable collaborative modeling among banks, core enterprises, logistics platforms, and other financial institutions while preserving commercial confidentiality and financial data privacy. This approach helps mitigate data silos in supply chain finance and supports more secure cross-institutional risk prediction.

Finally, at the level of practical application, achieving deep integration between the proposed research model and actual financial business processes is an important direction

for promoting the practical implementation of the research findings. Supply chain finance risk management depends not only on model prediction results but also on the coordinated operation of multiple processes, including credit approval, risk pricing, credit line adjustment, and post-loan monitoring. In this regard, the proposed framework can provide decision support for financial institutions in identifying high-risk SMEs, optimizing green credit allocation, and improving differentiated risk pricing. Future research may combine financial technology platform development with the embedding of risk prediction models into digital risk control systems, thereby constructing real-time risk warning and dynamic scoring mechanisms for specific business scenarios. Such integration can enable continuous monitoring of supply chain finance risk and further support the development of more transparent, data-driven, and green-oriented supply chain finance practices.

6. Conclusions

In summary, this study constructs a green supply chain finance credit risk indicator system using data from listed firms' annual reports, publicly available financial databases, supply chain disclosure information, and regional statistical materials. These data sources enhance the feasibility and traceability of the proposed framework and enable financial institutions to conduct preliminary risk profiling and credit risk screening of SMEs more efficiently. Building on this indicator system, the study further develops a risk prediction framework capable of extracting informative patterns from high-dimensional heterogeneous data and capturing complex nonlinear relationships and variable interactions in supply chain finance credit risk assessment.

The main research work and key findings of this study can be summarized as follows.

First, this study constructs a green supply chain finance credit risk evaluation system for SMEs in the automotive industry. Unlike traditional evaluation frameworks that rely mainly on firm-level financial indicators, the proposed system incorporates multidimensional information, including financial conditions, supply chain environment stability, core enterprise characteristics, green supply chain management, and digital transformation.

Second, this study applies the DE-LightGBM method to identify key risk features and reduce the influence of redundant and noisy variables. On this basis, TabPFGen is introduced to augment minority risk-class samples. By incorporating business-logic constraints and the NNDR filtering mechanism, the generated samples are made more consistent with the distributional and economic characteristics of real risk samples.

Third, the empirical results show that the TabPFN model performs effectively in identifying green supply chain finance credit risk among SMEs. Compared with XGBoost, CatBoost, LightGBM, logistic regression, and random forest, TabPFN achieves superior overall point-estimate performance in terms of accuracy, risk-class F1-score, and AUC. After data augmentation, the model exhibits further improvement in risk-class F1-score, Recall, and AUC, suggesting that TabPFGen-based minority-class augmentation can enhance TabPFN's ability to identify risky firms in small-sample and imbalanced tabular credit risk prediction tasks.

Fourth, the SHAP interpretation results reveal that total liabilities, supply chain environment stability, green supply chain management, current asset turnover, and the digital transformation index are among the key factors influencing the model's predictions. These findings indicate that credit risk in the context of green supply chain finance is not determined by a single financial dimension, but rather by the joint effects of capital strength, debt burden, operational efficiency, supply chain stability, and the level of green and digital management. The interpretability analysis therefore provides evidence on the underlying mechanisms through which financial, supply chain, green, and digital factors jointly affect SME credit risk.

Finally, this study further examines the incremental predictive value of green-related features. The results show that incorporating green-related information improves overall discriminative performance and risk-class F1-score, confirming that green performance and green supply chain management indicators provide additional explanatory and predictive information beyond conventional financial and supply chain variables.

Overall, under the constraints of limited data availability, this study constructs a comprehensive evaluation indicator system and verifies the effectiveness of the TabPFN-based prediction framework, as well as the incremental risk-identification value of green-related features. Nevertheless, this research remains a preliminary exploration based on real-world data constraints. Future research can further extend the sample to non-listed SMEs, additional industries, different national contexts, and economic environments, as well as transaction-level supply chain finance records, to improve the external validity of the proposed framework and examine its applicability across more heterogeneous business scenarios.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Table A1. Credit Risk Indicator System for Automotive SMEs in the Context of Green Supply Chain Finance.

First-Level Indicator	Second-Level Indicator	Third-Level Indicator
Macro-level Indicators	Manufacturing Business Climate	Manufacturing Purchasing Managers' Index
	Industry Environment	Automobile Production
		Year-on-year Growth Rate of Automobile Production
Comprehensive Indicators of the Financing Enterprise	Solvency	Current Ratio
		Quick Ratio
		Cash Ratio
		Debt-to-Asset Ratio
		Debt-to-Equity Ratio
		Interest Coverage Ratio
	Operating Efficiency	Fixed Asset Turnover
		Current Asset Turnover
		Total Asset Turnover

Table A1. *Cont.*

First-Level Indicator	Second-Level Indicator	Third-Level Indicator
Comprehensive Indicators of the Financing Enterprise	Profitability	Operating Profit Margin
		Return on Equity
	Growth Capacity	Growth Rate of Total Assets
		Growth Rate of Shareholders' Equity
		Growth Rate of Net Profit
		Growth Rate of Operating Revenue
		Technological Innovation Capability
	Firm Size and Capital Strength	Total Assets
		Registered Capital
	Composite Risk Early-warning Indicator	Z-score
	Green Performance	Wind ESG Rating
		Wind ESG Composite Score
		Greenhouse Gas Emissions
Comprehensive Supply Chain Indicators	Supply Chain Cooperation Performance	Customer Concentration
		Supplier Concentration
		Supply Chain Concentration
	Relationship Intensity	Years of Cooperation
		Frequency of Cooperation
	Green Supply Chain Performance	Carbon Information Disclosure
		Green Supply Chain Management
		Green Transformation Disclosure
		Green Supply Chain Management Strategy
		Implementation of Green Supplier Management
		Green Production
		Green Products
	Digitalization Level	Digitalization Outcome Score
		Digital Application Score
		Digital Transformation Index
	Collateral Characteristics	Inventory Quality
		Accounts Receivable Quality
		Prepayment Quality
	Trade Credit and Bargaining Power	Net Trade Credit Support Intensity
		Stability of the Supply Chain Environment
		Overall Bargaining Power

Table A1. *Cont.*

First-Level Indicator	Second-Level Indicator	Third-Level Indicator
Credit Support Capacity of Core Enterprises	Solvency	Current Ratio of the Core Enterprise
		Quick Ratio of the Core Enterprise
		Debt-to-Asset Ratio of the Core Enterprise
	Profitability	Operating Profit Margin of the Core Enterprise
		Return on Equity of the Core Enterprise
	Operating Efficiency	Accounts Receivable Turnover of the Core Enterprise
		Inventory Turnover of the Core Enterprise
	Growth Capacity	Growth Rate of Operating Revenue of the Core Enterprise
	Repayment Willingness	Nature of the Core Enterprise
		Core Enterprise Credit Rating

Table A2. Post-generation business constraints for TabPFGen-generated samples.

Constraint Category	Features Subject to Constraint	Processing Rule
Numerical range constraint	All normalized numerical features	Generated values were clipped to the valid [0, 1] interval to ensure consistency with the normalized feature scale.
One-hot categorical consistency constraint	Categorical features, including Wind ESG Rating, Green Supply Chain Management, Nature of the Core Enterprise, and Core Enterprise Credit Rating	For each one-hot encoded categorical group, only the category with the largest generated value was retained as 1, while all remaining categories were set to 0.
Admissible-value mapping for green-practice features	Green Supply Chain Management Strategy, Implementation of Green Supplier Management, Green Production, and Green Products	Generated values were mapped to the nearest admissible 0/1 values observed in the risk-class observations of the training set.
Green supply chain practice combination projection	Green Supply Chain Management Strategy, Implementation of Green Supplier Management, Green Production, and Green Products	Generated combinations not observed in the risk-class training observations were projected to the nearest observed green-practice combination.
Macro-industry combination projection	Manufacturing Purchasing Managers' Index, Automobile Production, and Year-on-year Growth Rate of Automobile Production	The three macro-industry features were treated as a joint annual combination and projected to observed training-set macro combinations.
Debt-structure combination projection	Interest-bearing Debt, Total Liabilities, and Interest-bearing Debt Ratio	If all three features were included in the final feature subset, generated debt-structure combinations were projected to the observed debt-structure combinations in the risk-class training observations.
General continuous feature treatment	Financial ratios, operating-efficiency indicators, digitalization-level indicators, green disclosure variables, collateral-quality indicators, and other continuous features	No additional subjective hard thresholds or cross-feature projection rules were imposed. These features were only subject to the [0, 1] range constraint and were subsequently included in NNDR-based quality filtering.

Note: All post-generation constraints were constructed using only the training set. The table summarizes the business-rule constraints applied before NNDR-based quality filtering, including numerical range clipping, one-hot categorical correction, admissible-value mapping, and combination projection.

Table A3. Distributional Similarity Test of Continuous Variables between Real and Synthetic Samples.

Variable	Real_Mean	Synthetic_Mean	Real_Std	Synthetic_Std	KS_Statistic	KSComplement	*p*_Value
Current Ratio	0.13	0.11	0.13	0.11	0.07	0.93	0.92
Quick Ratio	0.10	0.09	0.11	0.09	0.10	0.90	0.51
Debt-to-Asset Ratio	0.58	0.59	0.27	0.24	0.08	0.92	0.73
Interest Coverage Ratio	0.04	0.04	0.01	0.00	0.09	0.91	0.61
Fixed Asset Turnover	0.13	0.14	0.22	0.22	0.07	0.94	0.94
Current Asset Turnover	0.28	0.29	0.22	0.20	0.09	0.91	0.68
Total Asset Turnover	0.21	0.21	0.21	0.18	0.10	0.91	0.59
Return on Equity	0.65	0.69	0.20	0.17	0.11	0.89	0.46
Growth Rate of Net Profit	0.77	0.79	0.14	0.08	0.10	0.90	0.55
Z-score	0.14	0.13	0.11	0.07	0.09	0.91	0.69
Registered Capital	0.26	0.24	0.30	0.25	0.10	0.90	0.51
Net Trade Credit Support Intensity	0.51	0.52	0.27	0.25	0.07	0.93	0.90
Stability of the Supply Chain Environment	0.23	0.24	0.18	0.18	0.06	0.94	0.95
Digital Trans-formation Index	0.69	0.71	0.25	0.21	0.06	0.94	0.98
Supplier Concentration	0.31	0.28	0.21	0.18	0.08	0.92	0.73
Supply Chain Concentration	0.36	0.32	0.22	0.19	0.14	0.86	0.16
Current Ratio of the Core Enterprise	0.40	0.38	0.28	0.28	0.10	0.90	0.53
Debt-to-Asset Ratio of the Core Enterprise	0.64	0.65	0.26	0.25	0.09	0.92	0.72
Operating Profit Margin of the Core Enterprise	0.64	0.68	0.13	0.14	0.12	0.88	0.30
Accounts Receivable Turnover of the Core Enterprise	0.13	0.12	0.17	0.13	0.08	0.92	0.82
Inventory Turnover of the Core Enterprise	0.33	0.30	0.21	0.18	0.12	0.88	0.34
Total Liabilities	0.28	0.30	0.29	0.28	0.07	0.93	0.91
Interest-bearing Debt Ratio	0.49	0.53	0.25	0.23	0.07	0.93	0.92

Table A4. Distributional Similarity Test of Categorical Variables between Real and Synthetic Samples.

Variable	TVDistance	TVComplement
Green Supply Chain Management_Yes	0.29	0.71
Green Transformation Disclosure	0.12	0.89
Implementation of Green Supplier Management	0.02	0.98
Green Production	0.02	0.98
Green Products	0.05	0.95
Nature of the Core Enterprise_Central State-owned Enterprise	0.01	0.99
Nature of the Core Enterprise_Local State-owned Enterprise	0.02	0.98
Nature of the Core Enterprise_Private Enterprise	0.02	0.98

Table A5. Test-set performance of TabPFN models with 95% confidence intervals.

Model	F1 Score (Y = 1)	Precision (Y = 1)	Recall (Y = 1)	AUC	AUC-PR
Selected	0.8571 [0.7317, 0.9545]	0.9000 [0.6990, 0.9721]	0.8182 [0.6148, 0.9269]	0.9877 [0.9710, 0.9983]	0.9295 [0.8508, 0.9870]
Augmented	0.8837 [0.8077, 0.9778]	0.9048 [0.7109, 0.9735]	0.8636 [0.6667, 0.9525]	0.9886 [0.9717, 0.9990]	0.9413 [0.8706, 0.9924]

Note: Brackets report 95% confidence intervals. CIs for F1-score, AUC, and AUC-PR were obtained from 5000 stratified bootstrap resamples; CIs for Precision and Recall were calculated using the Wilson method.

References

1. Sun, R.; Chang, C.; Zhang, F.; Song, C.; Wang, C.; Li, X. Review and prospects of ESG practices in Chinese mining enterprises. *China Min. Mag.* **2025**, *34*, 15–24. [CrossRef]
2. Fan, C.; Peng, K. How does supply chain finance drive corporate green transition? The role of credit resource availability. *Financ. Res. Lett.* **2025**, *86*, 108381. [CrossRef]
3. Judijanto, L.; Utami, E.Y.; Harsono, I. Green supply chain finance: A bibliometric review of financing instruments, challenges, and opportunities. *West Sci. Interdiscip. Stud.* **2024**, *2*, 647–655. [CrossRef]
4. Zhu, Q.; Sarkis, J.; Lai, K.-H. Green supply chain management implications for "closing the loop". *Transp. Res. Part E Logist. Transp. Rev.* **2008**, *44*, 1–18. [CrossRef]
5. Yisa, J.; Taiwen, F. A literature review and prospects of green supply chain integration. *Foreign Econ. Manag.* **2022**, *44*, 135–152.
6. Yang, X. Research on risk assessment of green supply chain finance—A comparative study based on logit model and BP neural network. *J. Ind. Technol. Econ.* **2020**, *39*, 8.
7. Pfeifer, M.R. SMEs in automotive supply chains: A survey on six sigma performance perceptions of Czech supply chain members. *Processes* **2022**, *10*, 698. [CrossRef]
8. Song, H.; Yang, X.; Yu, K. How do supply chain network and SMEs' operational capabilities enhance working capital financing? An integrative signaling view. *Int. J. Prod. Econ.* **2020**, *220*, 107447. [CrossRef]
9. Zhu, Y.; Jia, R.; Wang, G.; Xie, C. A review of supply chain finance risk assessment research: Based on knowledge graph technology. *Syst. Eng.-Theory Pract.* **2023**, *43*, 795–812.
10. Akerlof, G.A. The market for "lemons": Quality uncertainty and the market mechanism. In *Uncertainty in Economics*; Elsevier: Amsterdam, The Netherlands, 1978; pp. 235–251.
11. Spence, M. Job market signaling. In *Uncertainty in Economics*; Elsevier: Amsterdam, The Netherlands, 1978; pp. 281–306.
12. Barney, J. Firm resources and sustained competitive advantage. *J. Manag.* **1991**, *17*, 99–120. [CrossRef]
13. Zhao, J.F.; Li, B. Credit risk assessment of small and medium-sized enterprises in supply chain finance based on SVM and BP neural network. *Neural Comput. Appl.* **2022**, *34*, 12467–12478. [CrossRef]
14. Li, G.; Yang, J.; Huang, Y. Supply chain finance credit risk evolving intelligent analysis system based on system dynamic model. *J. Intell. Fuzzy Syst.* **2020**, *38*, 7837–7847. [CrossRef]
15. Huang, X.; Sun, J.; Zhao, X. Credit risk assessment of supply chain financing with a grey correlation model: An empirical study on China's home appliance industry. *Complexity* **2021**, *2021*, 9981019. [CrossRef]
16. Huang, Y.; Lian, X. Identification and Assessment of Supply Chain Financial Risk in Core Enterprises Based on Telecommunications Industry. *J. Beijing Univ. Posts Telecommun.* **2016**, *18*, 44.
17. Mou, W.; Wong, W.-K.; McAleer, M. Financial credit risk evaluation based on core enterprise supply chains. *Sustainability* **2018**, *10*, 3699. [CrossRef]

18. Dai, X.; Yang, S. Digital empowerment, source of digital input and green manufacturing. *China Ind. Econ.* **2022**, *9*, 83–101.
19. Kou, D.; Zhang, C.; Zhang, X. Enterprise Digitalization Empowers Enterprise Green Transformation: From Passive Supervision to Active Innovation. *J. Beijing Univ. Technol. (Soc. Sci. Ed.)* **2024**, *24*, 48–66.
20. Liao, Z.; Zhang, M. Carbon finance, green supply chain management, and firms' environmental innovation: The moderating role of financing structure. *Int. J. Prod. Res.* **2025**, *63*, 8178–8190.
21. Sang, B. Application of genetic algorithm and BP neural network in supply chain finance under information sharing. *J. Comput. Appl. Math.* **2020**, *384*, 113170.
22. Zhang, H.; Shi, Y.; Yang, X.; Zhou, R. A firefly algorithm modified support vector machine for the credit risk assessment of supply chain finance. *Res. Int. Bus. Financ.* **2021**, *58*, 101482. [CrossRef]
23. Zhu, Y.; Xie, C.; Sun, B.; Wang, G.-J.; Yan, X.-G. Predicting China's SME credit risk in supply chain financing by logistic regression, artificial neural network and hybrid models. *Sustainability* **2016**, *8*, 433. [CrossRef]
24. Zhu, Y.; Zhou, L.; Xie, C.; Wang, G.-J.; Nguyen, T.V. Forecasting SMEs' credit risk in supply chain finance with an enhanced hybrid ensemble machine learning approach. *Int. J. Prod. Econ.* **2019**, *211*, 22–33. [CrossRef]
25. Hou, L.; Bi, G.; Guo, Q. An improved sparrow search algorithm optimized LightGBM approach for credit risk prediction of SMEs in supply chain finance. *J. Comput. Appl. Math.* **2025**, *454*, 116197. [CrossRef]
26. Hu, Z. Statistical optimization of supply chain financial credit based on deep learning and fuzzy algorithm. *J. Intell. Fuzzy Syst.* **2020**, *38*, 7191–7202. [CrossRef]
27. Xu, R.-Z.; He, M.-K. Application of deep learning neural network in online supply chain financial credit risk assessment. In *Proceedings of the 2020 International Conference on Computer Information and Big Data Applications (CIBDA), Guiyang, China, 17–19 April 2020*; IEEE: Piscataway, NJ, USA, 2020; pp. 224–232.
28. Tang, Q.; Lu, Y.; Wang, B.; Li, Z. A Deep Convolutional Neural Network Based Risk Identification Method for E-Commerce Supply Chain Finance. *Sci. Program.* **2022**, *2022*, 6298248. [CrossRef]
29. Mojdehi, K.F.; Amiri, B.; Haddadi, A. A novel hybrid model for credit risk assessment of supply chain finance based on topological data analysis and graph neural network. *IEEE Access* **2025**, *13*, 13101–13127. [CrossRef]
30. Nalule, D.; Edekebon, E.; Nagwovuma, M.; Jjingo, D.; Marvin, G. Explainable Deep Learning Approaches to Credit Risk Evaluation. In Proceedings of the 2024 Sixteenth International Conference on Contemporary Computing, Noida, India, 8–10 August 2024; pp. 198–206.
31. Zhang, W.; Yan, S.; Li, J.; Tian, X.; Yoshida, T. Credit risk prediction of SMEs in supply chain finance by fusing demographic and behavioral data. *Transp. Res. Part E Logist. Transp. Rev.* **2022**, *158*, 102611. [CrossRef]
32. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, Ł.; Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems*; Curran Associates, Inc.: Red Hook, NY, USA, 2017; Volume 30.
33. Hollmann, N.; Müller, S.; Eggensperger, K.; Hutter, F. Tabpfn: A transformer that solves small tabular classification problems in a second. *arXiv* **2022**, arXiv:2207.01848.
34. Chen, N.; Wei, Z.; Jin, X.; Lin, N.; Yang, F.; Zhao, L.; Wu, S. Integrating transformer-based learning and Sentinel-2 bare soil composites for soil organic carbon mapping in the black soil region of Northeast China. *Sci. Rep.* **2026**, *16*, 3784. [CrossRef] [PubMed]
35. Shang, B.; Li, G.; Sun, W.; Zhang, L.; Cui, G.; Tu, J.; Fang, X.; Li, X. In-context learning for nano-PCM thermal behavior prediction in battery thermal management via Lattice Boltzmann simulation. *Energy* **2025**, *338*, 138693. [CrossRef]
36. Li, Z.; Xiong, C.; Zheng, K.; Li, Q. An RF-TabPFN-Based Framework for Few-Shot IoT Network Attack Recognition Using Lasso-RFE Feature Selection. *IEEE Access* **2025**, *13*, 151452–151465. [CrossRef]
37. Xin, Z.-C.; Zhang, J.-S.; Lan, M.; Liu, Q. TabPFN-SHAP-based slag viscosity prediction model with high accuracy, efficiency, and interpretability. *Metall. Mater. Trans. B* **2025**, *56*, 4330–4336. [CrossRef]
38. Ma, J.; Dankar, A.; Stein, G.; Yu, G.; Caterini, A. TabPFGen--Tabular Data Generation with TabPFN. *arXiv* **2024**, arXiv:2406.05216.
39. Tugnoli, D.; De Lorenzo, A.; Virgolin, M.; Cinà, G. Improving TabPFN's Synthetic Data Generation by Integrating Causal Structure. *arXiv* **2026**, arXiv:2603.10254.
40. Liu, Y.; Li, S.; Yu, C.; Lv, M. Research on green supply chain finance risk identification based on two-stage deep learning. *Oper. Res. Perspect.* **2024**, *13*, 100311. [CrossRef]
41. Liu, J.; Liu, S.M.; Li, J.; Li, J.Z. Financial credit risk assessment of online supply chain in construction industry with a hybrid model chain. *Int. J. Intell. Syst.* **2022**, *37*, 8790–8813. [CrossRef]
42. Belhadi, A.; Kamble, S.S.; Mani, V.; Benkhati, I.; Touriki, F.E. An ensemble machine learning approach for forecasting credit risk of agricultural SMEs' investments in agriculture 4.0 through supply chain finance. *Ann. Oper. Res.* **2025**, *345*, 779–807. [CrossRef]
43. Yuwono, M.A.; Rachmawati, D. The Role of Risk Management in Improving the Operational Effectiveness of SMEs: A Systematic Review. *JISR Manag. Soc. Sci. Econ.* **2024**, *22*, 1–25. [CrossRef]

44. Shan, W.; Gao, B. Stacked Ensemble Model with Enhanced TabNet for SME Supply Chain Financial Risk Prediction. *Systems* **2025**, *13*, 892. [CrossRef]
45. Yang, Y.; Chu, X.; Pang, R.; Liu, F.; Yang, P. Identifying and predicting the credit risk of small and medium-sized enterprises in sustainable supply chain finance: Evidence from China. *Sustainability* **2021**, *13*, 5714. [CrossRef]

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